

# ORTPiece: An ORT-Based Turkish Image Captioning Network Based on Transformers and WordPiece

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**Abstract**—Recent transformers-based systems are advancing image captioning applications. However, those works have been mainly applied to English-based image captioning problems. In this paper, we introduce a transformers-based Turkish-based image captioning algorithm. Our proposed algorithm uses appearance and geometry features from the input image and combines them along with the WordPiece embeddings to generate the Turkish-based caption. Our experimental results show improvement when compared to the other existing techniques including the original ORT and the show-and-tell algorithms.

**Keywords**—*Turkish Image captioning, Transformer, Object Relation Transformer, WordPiece Tokenization*

**Özetçe** —Son transformatör (dönüştürücü) tabanlı sistemler, görüntü altyazılama uygulamalarını geliştirmektedir. Ancak, bu çalışmalar esas olarak İngilizce tabanlı görüntü altyazılama problemlerine uygulanmıştır. Bu bildiride, transformatör kullanan Türkçe tabanlı bir görüntü altyazılama algoritması sunuyoruz. Önerilen algoritmamız, verilen görüntüdeki görünüm ve geometri özelliklerini kullanır ve bunları Türkçe tabanlı altyazı oluşturmak için WordPiece yerleştirmeleriyle birleştirir. Deneysel sonuçlarımız, ORT ve "göster ve anlat" (Show and tell) algoritmaları ile karşılaştırıldığında iyileştirme göstermektedir.

**Anahtar Kelimeler**—*Türkçe görüntü altyazılama; Dönüştürücü; Nesne ilişkilendiren dönüştürücü; WordPiece belirteç oluşturu.*

## I. INTRODUCTION

Image captioning is the task of generating a textual description of a given image by combining natural language processing techniques with computer vision. Recent advancements in deep learning improved many applications as in [8], [13], [18], as well as image captioning. A particular advancement in text processing is the introduction of the attention mechanism [22] which allows the model to pay attention to specific regions of the image when generating the caption. Geometric attention [7] is a type of attention mechanism used in image captioning, which weights different regions of an image based on their geometric properties such as position, size, and aspect ratio. Through geometric attention, [5] incorporated spatial relationship information between a pair of detected objects, which could be essential to understand the content of the image, by modifying the encoder layer of the standard transformer [22]. While image captioning remains an important computer vision task, its applications have been limited in the Turkish language. [20] introduced the first Turkish image captioning dataset. Later, additional translation-based datasets as in [19] and [26] were also proposed. Most of the existing work (as in

[9], [19], [26]) related to the Turkish-based image captioning relies on the show and tell's [24] model which consists of two main parts: a convolutional neural network (CNN) for image feature extraction, and a long short-term memory (LSTM) network for language generation, which is an old model. While LSTM-based techniques have been useful, recent advances showed that transformer-based solutions can provide improved performance in many applications.

In this paper, we introduce a transformer-based solution to a Turkish-based image captioning system. Our transformer-based deep architecture uses an encoder and a decoder based architecture. We improve upon Object Relation Transformer (ORT) [5] by using a WordPiece-based [3] model trained on the translated captions to tokenize the captions fed to the decoder instead of whitespace-based tokenized captions. This addition addresses the fact that Turkish is a highly agglutinative language, which means that it uses a large number of suffixes to create words and convey grammatical information. Our contributions include: (i) introduce a transformer-based architecture for Turkish image captioning; (ii) use a WordPiece-based model to improve performance and to analyze the effect of the vocab size on the performance; (iii) present state-of-the-art results when compared to the Show and tell, and ORT-based techniques; (iv) provide a set for training, validation, and testing of Turkish splits parallel to the English splits format, a Turkish gold captions file usable with the open-source evaluation tools made for English, and our implementation<sup>1</sup>.

## II. RELATED WORK

Deep learning-based approaches have been widely adopted by many applications including autonomous vehicles, computer vision [21] and image captioning. Recently proposed image captioning methods achieve significant progress in the field as in [24], [25], [2] and [5]. However, English was usually the language of interest due to the availability of English captions datasets [11], [15], [6]. The authors of [20] introduced TasvirEt which is the first Turkish image captioning dataset. TasvirEt was created by collecting Turkish captions for images in the Flickr8K [6] dataset. A system was developed and the captions were collected in a crowd-sourced manner. The data entry was designed to be publicly accessible where anyone could contribute. They utilized a nearest-neighbor based technique to build the first Turkish image captioning model. Later, the authors of [19] translated the captions of the COCO training

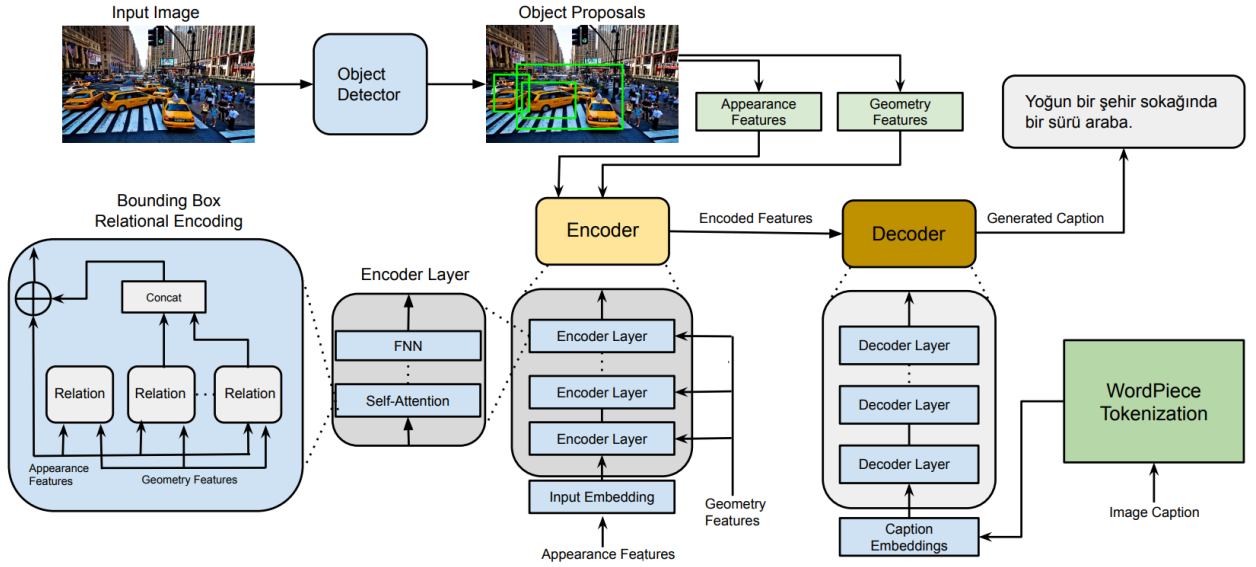


Figure 1: ORTPiece architecture which is based on the Object Relational Transformer (ORT) and the Wordpiece algorithms is illustrated. An object detector generates the appearance and geometry based features first, then those features along with the WordPiece embeddings are fed into the decoder block to generate the captions. The details of the encoder layer and self-attention layer are also provided in this figure.

set from English into Turkish by using Google’s Translate API. They trained an encoder-decoder model based on the model in [24] and manually captioned (by two people) 500 test images which were randomly selected from the validation set of the COCO dataset. The work in [9], was one of the earliest studies to address the fact that Turkish is an agglutinative language. They first applied a Byte-Pair Encoding (BPE) algorithm [17] on the captions to divide the words into subwords. BPE starts with a set of characters and repeatedly merges the most frequent pair of characters or subwords until a certain number of subwords is reached. The resulting subwords are usually not real words and are not guaranteed to have any semantic meaning, which may not be suitable for Turkish. As done in [19], the work in [26] translated the dataset again by using Yandex’s Translation API. The earlier mentioned Turkish image captioning papers including [19], [26], and [9] essentially utilized models that use *Show and Tell* [24] algorithm as basis to train a Turkish image captioning model. While Show and Tell is one of the pioneering works in this area and achieved state-of-the-art at the time, many recent models have since been proposed that outperform it. Our proposed model does not use Show and Tell as baseline; instead, it uses transformers and is based on the ORT [5] algorithm.

### III. OUR PROPOSED APPROACH: ORTPIECE

Our network uses an encoder-decoder architecture and its overview is provided in Figure 1. We name our solution captioning images in Turkish as ORTPiece. The input image is going through an object detector to obtain appearance and geometry-based features. Those features, then are given as input to the encoder architecture. The word-piece embedding and the encoder’s output are fed into the decoder block as input. The output of the decoder is the text-based image caption in Turkish. In our implementation, we use a pre-trained Faster R-CNN [16] as the detector, and train only the encoder and

decoder parts. Our encoder is based on the architecture from [5] which is a modified version of the encoder in the standard transformer model [22], and the decoder block is based on the decoder architecture as proposed in [24]. Figure 1 shows the overview of our Turkish image captioning pipeline, which uses the [5] model as its baseline and uses an additional *tokenization* component. The encoder modifies the attention weight matrix to add spatial relationship information between the objects. It combines the appearance-based attention weights with the geometric attention weights by multiplying the appearance-based attention weights of two objects by a learned function of their relative position and size. The encoder’s input is the features extracted by Faster R-CNN [16] with ResNet-101 [4]. Our decoder block is based on the decoder block from [22]. We use a WordPiece-based tokenization [3] model to preprocess the captions fed to the decoder.

**WordPiece Tokenization:** WordPiece tokenization [3] is a technique used in natural language processing. It involves breaking down a word into subword units called wordpieces. The algorithm starts with an initial vocabulary that contains all individual characters and subword units. It then determines the frequency of each subword, which helps in deciding which wordpieces to merge. With the initial dictionary and frequencies, the algorithm carries on, for iteratively merging the most frequent pairs of wordpieces until a predetermined vocabulary size is reached and the vocabulary is constructed. Since Turkish is a highly agglutinative language, WordPiece helps in breaking down words with long suffixes rather than treating them as out-of-vocabulary words.

### IV. EXPERIMENTS

Our implementation builds on top of the implementation of the work from [5]. We train our WordPiece model using



**Show and Tell:** Çimenli bir alanda duran bir grup kuş .  
**ORT:** Çimenli bir alanda duran bir grup kuş .  
**ORTPiece:** kuru ot alanının üstünde duran üç kuş .

Bir şehir sokağında toplu taşıma otobüsü  
 Bir şehir sokağında trafik ışığı ve sarı bir araba .  
 yoğun bir şehir sokağında bir sürü araba .

Figure 2: Generated captions of three algorithms (Show and Tell, ORT and ORTPiece) are shown for two randomly selected images from the COCOTurk’s test set.

| Dataset        | Number of Images |            |       | Captions per Image | Translated Set |
|----------------|------------------|------------|-------|--------------------|----------------|
|                | Train            | Validation | Test  |                    |                |
| Flickr8k [6]   | 6000             | 1000       | 1000  | 5 (English)        | None           |
| Flickr30k [15] | 31783            | -          | -     | 5 (English)        | Train          |
| COCO [11]      | 82783            | 40504      | 40775 | 5 (English)        | Train          |
| TasvirEt [20]  | 6000             | 1000       | 1000  | 2 (Turkish)        | -              |
| COCOTurk [19]  | 77783            | 5000       | 500   | 5-5-2 (Turkish)    | -              |

Table I: This table summarizes the statistics of various datasets used for image captioning.

HuggingFace’s tokenizers<sup>2</sup> (last accessed on May 2023). We use the same hyperparameters of [5] and the same loss function, cross-entropy, with some changes, while running our experiments. Those changes are listed as follows: (i) instead of training for 30 epochs, we train our models for only 20 epochs, which seemed more reasonable in this case as the loss stops decreasing beyond that in preliminary experiments; (ii) we also evaluate our models at the end of every epoch rather than after a specified number of iterations.

**Dataset:** We use the COCO translated dataset from [19]. We randomly sampled 5000 images from the training set to form the validation set. Our test set is composed of 500 images and was proposed by [19]. It has two captions per image. Table II summarizes the relevant image captioning datasets and their statistics. We utilize [19] and format it following *Karpathy*’s splits format, and we call it COCOTurk throughout this paper.

**Metrics:** We evaluate our models on COCOTurk’s test set [19]. We report our results on several metrics, which are commonly used for image captioning. Some of those metrics are borrowed from other NLP tasks. For example, Bilingual Evaluation Understudy (BLEU) [14] is a metric for evaluating the quality of text which has been machine-translated from one natural language to another. It is calculated as the geometric mean of the n-gram precisions of the number of n-grams shared between the original and the output text. Metric for Evaluation of Translation with Explicit Ordering (METEOR) [12], on the other hand, is calculated as the harmonic mean of the precision and recall of the n-grams shared between the original and the output text. They are both

| Model/Metric       | Vocab Size | CIDEr-D      | SPICE        | BLEU-1       | BLEU-4       | METEOR       | ROUGE-L      |
|--------------------|------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Show and Tell [24] | 19k        | 0.408        | 0.076        | 0.297        | 0.033        | 0.147        | 0.305        |
| Show and Tell [24] | 24k        | 0.409        | 0.069        | 0.295        | 0.036        | 0.144        | 0.298        |
| Show and Tell [24] | 39k        | 0.448        | 0.076        | 0.308        | 0.039        | 0.151        | 0.298        |
| Topdown [2]        | 24k        | 0.508        | 0.073        | 0.319        | 0.044        | 0.168        | 0.317        |
| Transformer [22]   | 24k        | 0.477        | 0.073        | 0.313        | 0.039        | 0.159        | 0.316        |
| ORT [5]            | 19k        | 0.523        | 0.075        | 0.328        | 0.054        | 0.171        | 0.327        |
| ORT [5]            | 24k        | 0.526        | 0.078        | <b>0.334</b> | 0.052        | 0.173        | 0.325        |
| ORT [5]            | 39k        | 0.513        | <b>0.079</b> | 0.321        | 0.050        | 0.167        | 0.324        |
| ORTPiece           | 19k        | 0.526        | 0.077        | 0.330        | <b>0.055</b> | 0.172        | 0.325        |
| ORTPiece           | 24k        | <b>0.553</b> | 0.077        | 0.333        | 0.053        | <b>0.174</b> | <b>0.329</b> |
| ORTPiece           | 39k        | 0.524        | 0.074        | 0.331        | 0.053        | 0.172        | 0.326        |

Table II: Experimental results are shown in various metrics on the COCOTurk’s test set. Best results are shown in bold.

usually used in evaluating machine translation systems. Recall-Oriented Understudy for Gisting Evaluation (ROUGE-L) [10] is a metric used for summarization and machine translation tasks. It measures the longest common subsequence (LCS) between the automatically produced text and the original text. Additionally, specific metrics for image captioning, such as the Consensus-based Image Description Evaluation (CIDEr) [23] score, which is a metric used to evaluate the quality of image captions systems. It measures the similarity between the generated captions and human reference captions. CIDEr addresses the limitations of BLEU, which primarily focuses on n-gram overlap. It takes into account not only the word-level overlap between the input and the output captions but also the consensus among human annotators, which is two in our case. The algorithm assigns a weight to each n-gram based on its rarity and consensus. It then calculates the similarity score by comparing the frequency of each n-gram in the generated caption with its corresponding weight and finally combines the weighted similarity scores for all n-grams. Finally, Semantic Propositional Image Caption Evaluation (SPICE) [1] is a metric developed for the image caption problem, which evaluates the captions based on five semantic categories: object, attribute, relationship, spatial, and comparative. It analyzes how well the generated captions capture the relevant semantic categories. We ran several experiments with the purpose of comparing ORT [5] with our ORTPiece model, as well as with older models [2], [24] which were used in the literature for Turkish

<sup>2</sup><https://github.com/huggingface/tokenizers>

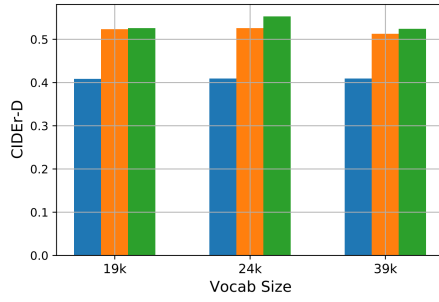


Figure 3: This "vocabulary (Vocab) size" versus the "CIDEr-D score" plot compares the experimental results of algorithms including Show and Tell (in blue), ORT (in orange), and ORTPiece (in green) for three different vocabulary sizes.

image captioning such as [9], [19], [26]. For a fair comparison, we used the same data splits, the same hyperparameters, and the same experimental setup for each experiment. The results obtained on the COCOturk's test set [19] are shown in Table II. We test the same model with different vocabulary (vocab) sizes to analyze the effect of the vocab size and also to compare the models fairly. 2 shows ORTPiece outperforming all the other models with a CIDEr-D score of 0.553. ORT [5], on the other hand, has a score of 0.079 and 0.334 for SPICE and BLEU-1 scores, respectively, achieving higher scores than the other models. For BLEU-4, METEOR, and ROUGE-L scores, ORTPiece again surpasses the other models with scores of 0.005, 0.174, and 0.0329 respectively. To understand the effect of the vocab size on the models, we train two models: [5], [24], along with ORTPiece, with three different vocab sizes: 19k, 24k, and 39k. Figure 3 shows the results of those three models with a vocab size of 24k achieve a higher CIDEr-D score than those with the other vocab sizes.

## V. CONCLUSION

In this paper, we proposed a transformers-based novel approach for Turkish-based image captioning of a given input image. Our algorithm builds on the ORT algorithm and uses WordPieces embeddings. The output is a text-based description (caption) in Turkish. Our architecture combines WordPiece embeddings along with appearance and geometry features for captioning. Our preliminary results on COCOturk dataset demonstrate that our proposed approach (called ORTPiece) yields the state of the art results when compared to its baseline network: ORT and when it is compared to the show-and-tell network for Turkish based image captioning.

## REFERENCES

- [1] Peter Anderson, Basura Fernando, Mark Johnson, and Stephen Gould. Spice: Semantic propositional image caption evaluation. pages 382–398, 2016.
- [2] Peter Anderson, Xiaodong He, Chris Buehler, Damien Teney, Mark Johnson, Stephen Gould, and Lei Zhang. Bottom-up and top-down attention for image captioning and visual question answering. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6077–6086, 2018.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. pages 770–778, 2016.

- [5] Simao Herdade, Armin Kappeler, Kofi Boakye, and Joao Soares. Image captioning: Transforming objects into words. *Advances in neural information processing systems*, 32, 2019.
- [6] Micah Hodosh, Peter Young, and Julia Hockenmaier. Framing image description as a ranking task: Data, models and evaluation metrics. *Journal of Artificial Intelligence Research*, 47:853–899, 2013.
- [7] Han Hu, Jiayuan Gu, Zheng Zhang, Jifeng Dai, and Yichen Wei. Relation networks for object detection. pages 3588–3597, 2018.
- [8] Hüseyin Enes İlhan, Sedat Ozer, Güneş Karabulut Kurt, and Hakan Ali Cirpan. Offloading deep learning empowered image segmentation from uav to edge server. In *2021 44th International Conference on Telecommunications and Signal Processing (TSP)*, pages 296–300. IEEE, 2021.
- [9] Menekşe Kuyu, Aykut Erdem, and Erkut Erdem. Image captioning in turkish with subword units. pages 1–4, 2018.
- [10] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. 2004.
- [11] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. pages 740–755, 2014.
- [12] Alon Lavie Michael Denkowski. Meteor universal: Language specific translation evaluation for any target language. 2014.
- [13] Sedat Özer, Mert Ege, and Mehmet Akif Özkanoglu. Siamesefuse: A computationally efficient and a not-so-deep network to fuse visible and infrared images. *Pattern Recognition*, 129:108712, 2022.
- [14] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. pages 311–318, 2002.
- [15] Bryan A Plummer, Liwei Wang, Chris M Cervantes, Juan C Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *Proceedings of the IEEE international conference on computer vision*, pages 2641–2649, 2015.
- [16] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. 2016.
- [17] Alexandra Birch Rico Sennrich, Barry Haddow. Neural machine translation of rare words with subword units. 2016.
- [18] Oyku Sahin and Sedat Ozer. Yolodrone+: Improved yolo architecture for object detection in uav images. In *30th IEEE Signal Processing and Communications Applications Conference (SIU)*, pages 1–4, 2022.
- [19] Nermin Samet, Samet Hiçsönmez, Pınar Duygulu, and Emre Akbaş. Could we create a training set for image captioning using automatic translation? pages 1–4, 2017.
- [20] Mesut Erhan Unal, Begum Citamak, Semih Yagcioglu, Aykut Erdem, Erkut Erdem, Nazlı İkizler Cinbis, and Ruket Cakici. Tasviret: A benchmark dataset for automatic turkish description generation from images. pages 1977–1980, 2016.
- [21] Rodolfo Valiente, Mahdi Zaman, Yaser P Fallah, and Sedat Ozer. Connected and autonomous vehicles in the deep learning era: A case study on computer-guided steering. In *Handbook Of Pattern Recognition And Computer Vision*, pages 365–384. World Scientific, 2020.
- [22] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [23] Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. Cider: Consensus-based image description evaluation. pages 4566–4575, 2015.
- [24] Oriol Vinyals, Alexander Toshev, Samy Bengio, and Dumitru Erhan. Show and tell: A neural image caption generator. pages 3156–3164, 2015.
- [25] Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhudinov, Rich Zemel, and Yoshua Bengio. Show, attend and tell: Neural image caption generation with visual attention. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2048–2057, 2015.
- [26] Berk Dursun Yılmaz, Ali Emre Demir, Elena Battini Sönmez, and Tuğba Yıldız. Image captioning in turkish language. pages 1–5, 2019.